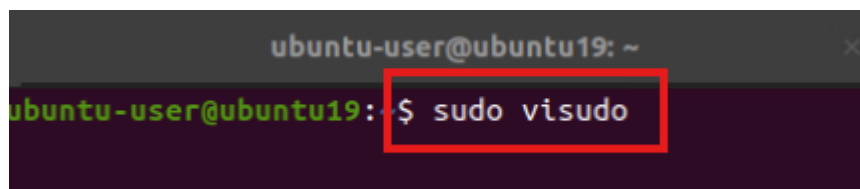


Sudo is a powerful tool that allows non-root users to execute commands as root. When set up correctly it can greatly increase the security of a system by restricting the set of operations a given user can do as root.

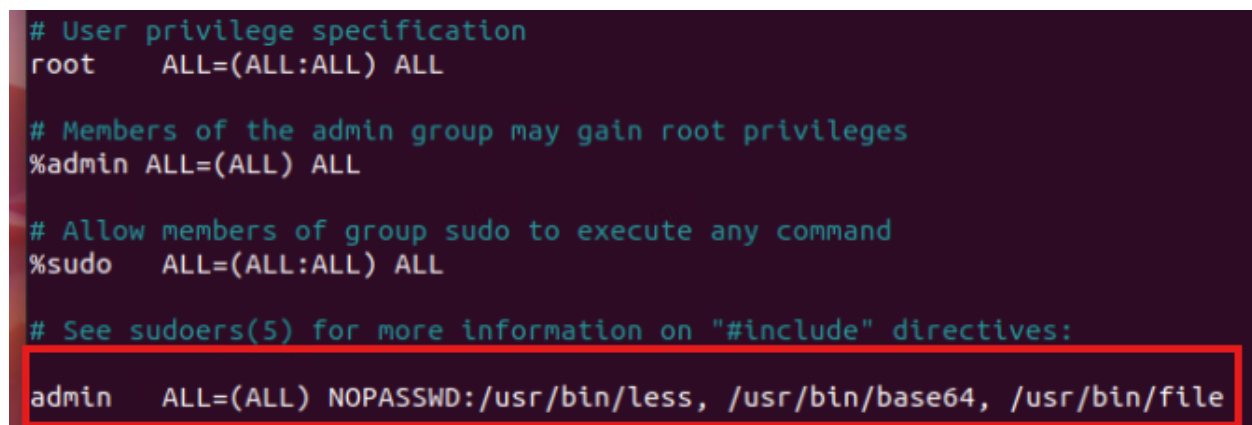
However, when not carefully set up sudo can also allow users to do far more than may have been intended.

Securing a sudo configuration will always be a matter of balancing restricting access and avoiding limiting a user's ability to do critical work. As always, the principle of least privilege should be followed. The sudo configuration file can be found at `/etc/sudoers`.

The sudo configuration file can be editing via `[sudo visudo]` command.

A terminal window with a dark background. The prompt is 'ubuntu-user@ubuntu19: ~'. The command '\$ sudo visudo' is entered and highlighted with a red rectangular box.

From this file you can see that the admin user can run any command as any user using sudo. Removing this "no password" rule is a good first step in securing this configuration as it allows the password to provide a layer of defense.

A terminal window showing the contents of the /etc/sudoers file. The text is as follows:
User privilege specification
root ALL=(ALL:ALL) ALL

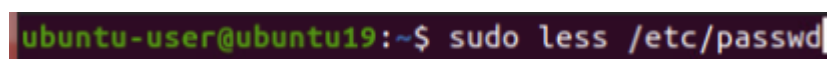
Members of the admin group may gain root privileges
%admin ALL=(ALL) ALL

Allow members of group sudo to execute any command
%sudo ALL=(ALL:ALL) ALL

See sudoers(5) for more information on "#include" directives:

admin ALL=(ALL) NOPASSWD:/usr/bin/less, /usr/bin/base64, /usr/bin/file
The last line is highlighted with a red rectangular box.

Lastly, we'll take a look at the program less. less allows for content that would otherwise fill the terminal screen to be paged. However, less also has a lesser-known ability to run commands. Leveraging this we will be able to enter a root shell.

A terminal window with a dark background. The prompt is 'ubuntu-user@ubuntu19:~'. The command '\$ sudo less /etc/passwd' is entered.

Once in the less screen, press '!', followed by the command you would like to execute. Since we would like to spawn a shell. [! /bin/sh]

```
saned:x:116:122:./var/lib/saned:/usr/sbin:/bin/sh
nm-openvpn:x:117:123:NetworkManager OpenVPN:/usr/sbin:/bin/sh
! /bin/sh
```

You should now be in a root shell.

```
# id
uid=0(root) gid=0(root) groups=0(root)
# whoami
root
#
```